



Strategic Partner



E&ICT Academy, IIT Kanpur

A Joint Initiative of MeitY & IIT Kanpur

Strategic Financial Governance & Leadership

SESSION: 5

MENTOR: Shubhashish Biswas

Session 5 Agenda

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5.1

Governance foundations for CTOs

5.2

The role of the CTO in strategic financial governance & decision-making

5.3

Implementing financial strategies in IT / tech function

5.4

Building a financially savvy technology team



5.1

Governance foundations for the CTO

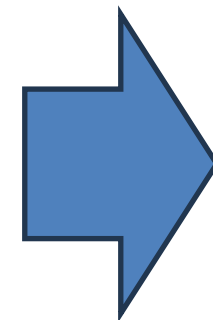
Introduction to Governance

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Control objects of the modern CTO:

- **Assets** — what we own, subscribe to, commit to, and consume
- **Vendors** — who powers our technology stack and under what terms
- **Resources** — how our internal and external delivery capacity is deployed



The finance question to ask

- **Assets**: Are we paying for what creates value?
- **Vendors**: Are we buying under the right commercial and risk terms?
- **Resources**: Are we putting the right talent against the right work?

Introduction to Governance

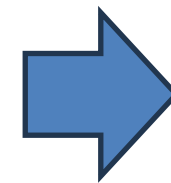
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“A weak governance model makes even a good budget look bad later.”

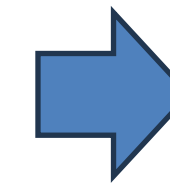
Reality

Enterprise IT is fragmented across SaaS, cloud, internal platforms, offline servers, software licenses, legacy systems, AI tools, vendors, and blended delivery teams. This creates room for multiple leakage points in the IT org.



Top Value Leakage points:

1. Unused licenses,
2. Wrong contract terms,
3. Overcommitted cloud capacity,
4. Unmanaged renewals,
5. Contractor sprawl,
6. Poor allocation of engineering effort



CTO's core meaning of governance:

The CTO's governance role is to prevent those **value leaks**.

Asset Governance

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Control objects of the modern CTO:

1.Assets — what we own, subscribe to, commit to, and consume

1.Vendors — who powers our technology stack and under what terms

1.Resources — how our internal and external delivery capacity is deployed

Content

- Hardware
- Perpetual software and licenses
- SaaS subscriptions
- Cloud reservations / savings plans / committed-use discounts
- Internal platforms and software products
- Data platform subscriptions
- AI model/service commitments
- Support and maintenance entitlements

Key Distinctions

- Not every asset is capitalized
- Not every contract is economically flexible
- Some “assets” are really usage commitments in disguise

Asset Lifecycle & Realities

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Lifecycle

Request → approve → procure → provision → assign owner → use → measure → renew/resize → retire

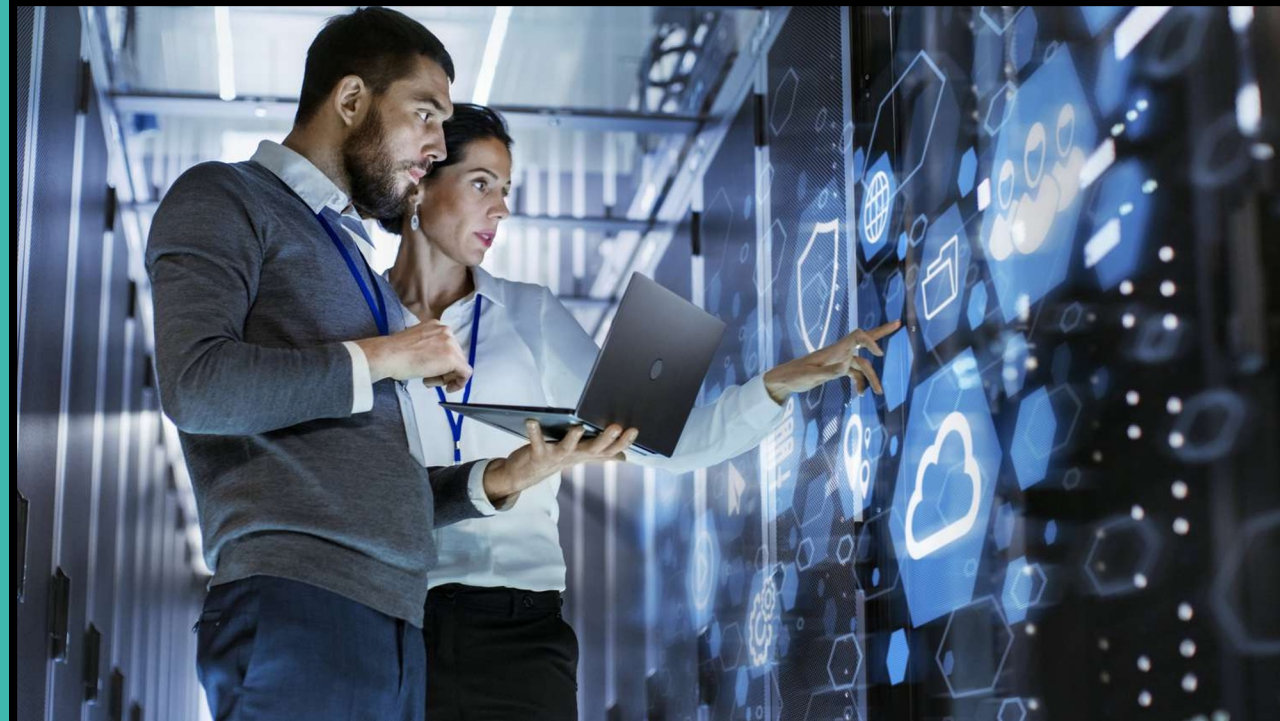
Most asset waste comes from one of five failures:

- no complete inventory,
- no named owner,
- no usage data,
- no contract linkage,
- no retirement discipline.

If an item has no business owner, no renewal date, or no utilization view, it is not governed — it is merely purchased.

Mini Case 1

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A firm buys **1,200 seats** of a collaboration/security/developer tool at **₹24,000 per user per year**.

Total annual spend = **₹2.88 crore**.

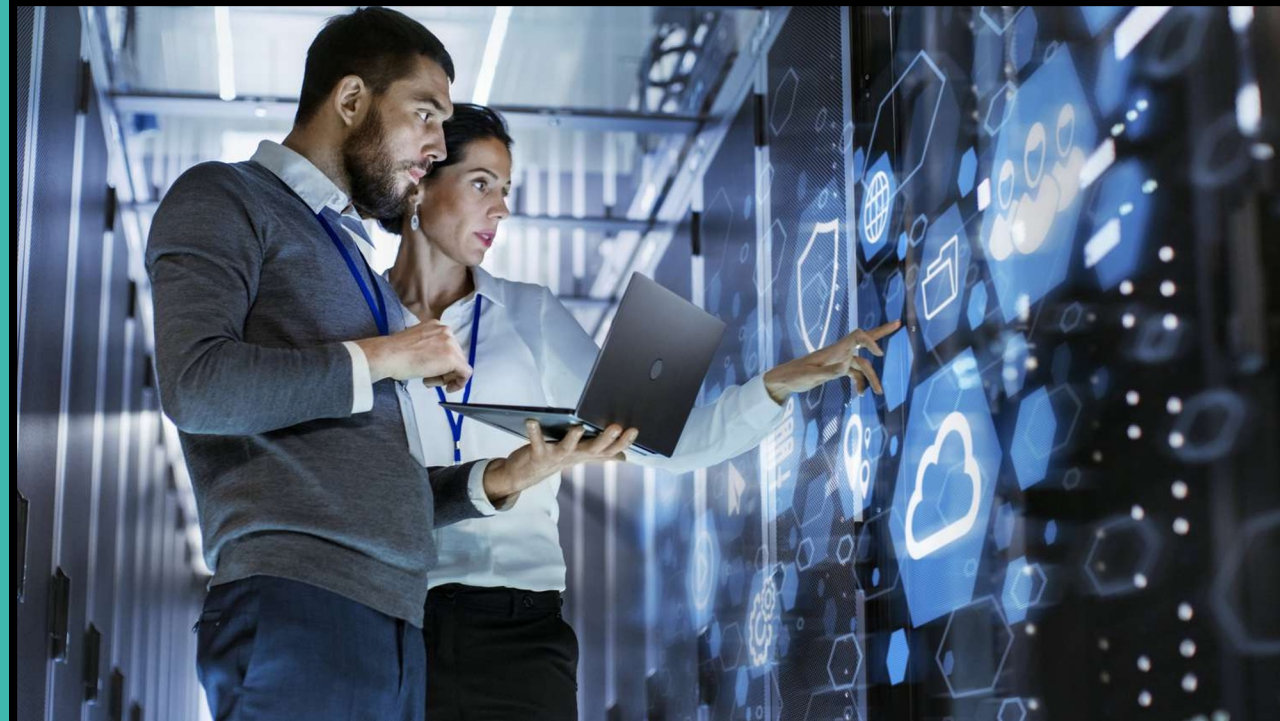
After 90 days of usage review, we find:

- 690 users are active monthly
- only 480 users use premium features
- 210 users left the company or moved roles
- another 300 users log in but do not need the premium tier

1. **What questions should the CTO ask after seeing this review status?**
2. **What action plan should be drawn after that? What options does the CTO have anyway?**

Mini Case 1

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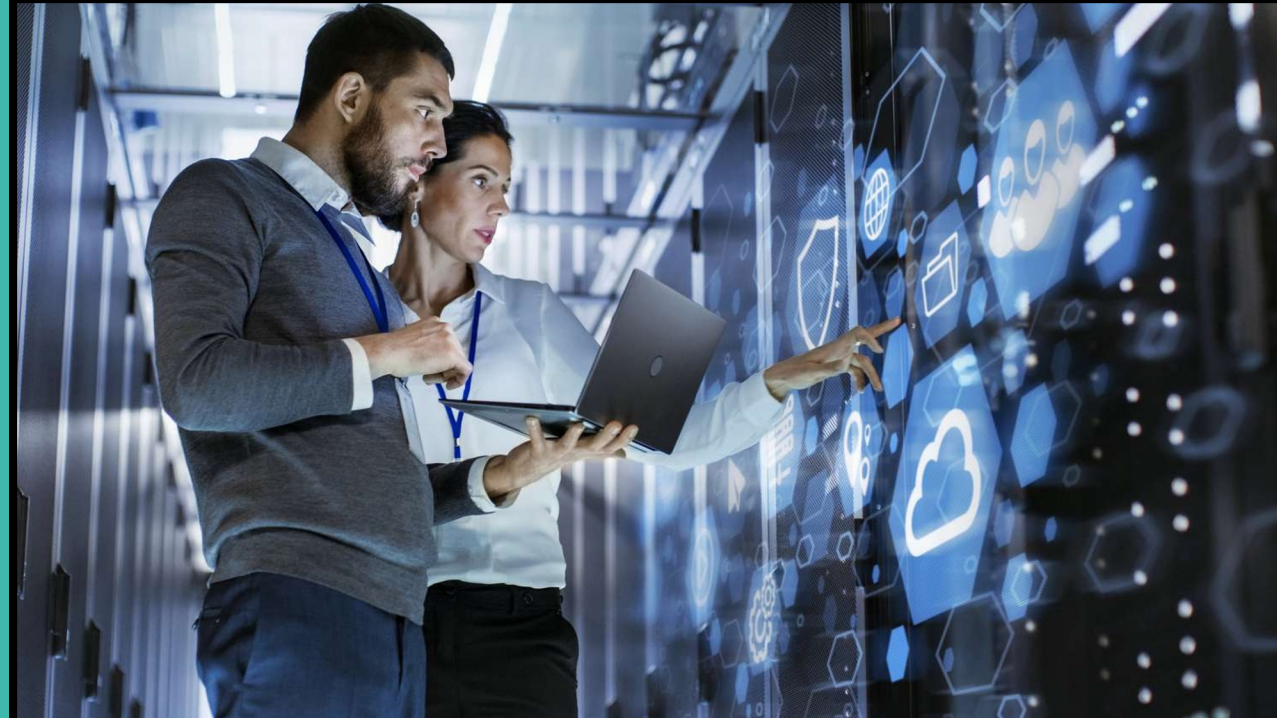
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CTO's core questions should be:

1. Are we managing by provisioned seats or active seats?
2. Do we need one tier or multiple tiers?
3. Do we reclaim licenses monthly or only at renewal?
4. Is the tool duplicated elsewhere?

Mini Case 1

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only 480 users use premium features

210 users left the company or moved roles

another 300 users log in but do not need the premium tier

Possible action path

1. reduce total seats,
2. downgrade feature tiers,
3. automate quarterly reclamation,
4. force manager re-approval for dormant seats.

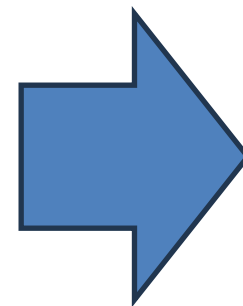
Cloud Based Financial Governance Needs

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Cloud commitment is risky, and needs steady governance

Cloud commitments can produce major savings, but only when the workload is genuinely stable.



Top OEMs' offers are alluring, which may cause governance oversights:

- AWS states Savings Plans can save up to 72% on eligible compute usage;
- Microsoft Azure Reservations similarly advertise savings up to 72% for qualifying one-year or three-year reservations;
- Google Cloud's committed-use discounts use one-year or three-year commitments, and its Compute Engine pricing page shows examples such as 28% for a 1-year commitment and 46% for a 3-year commitment over committed hourly spend.

Mini Case 2

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A company has a stable baseline compute need of ₹20 lakh/month. At on-demand rates:

Annual spend = ₹2.4 crore

If the team is very confident about stability, it can make a:

- a 1-year style commitment at ~28% savings reduces annual cost materially
- a 3-year style commitment at ~46% savings reduces cost further

- **What financial trap should the CTO foresee here?**
- **What action plan should be made to avoid the trap?**

Mini Case 2

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If the architecture changes, traffic falls, or workloads move, the firm may still be paying for committed usage it no longer needs.

Mini Case 2

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- a 1-year style commitment at ~28% savings reduces annual cost materially
- a 3-year style commitment at ~46% savings reduces cost further

What action plan should be made to avoid the trap?

1. Find out what portion of spend is stable enough to commit, for how long,
2. What are the migration uncertainties, and how many resource units fall in its purview?
3. Reassess the need, with a scenario analysis, with certainty levels measures for each.
4. Make commitments based on 100% certain parts, and 50-70% reduction on each decile of reduction in certainty.



5.2

The role of the CTO in strategic financial decision-making

A CTO's Scope of financial leadership

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THE 4 DIMENSIONS OF A CTO's SCOPE OF FINANCIAL LEADERSHIP

The role of the Chief Technology Officer (CTO) in strategic financial decision-making is multifaceted and crucial for aligning technology investments with the overall financial health and strategic goals of an organization.



Strategy,
Vision &
Alignment



Evaluation /
Prioritization



Risk
Management



Communication
& Assignment

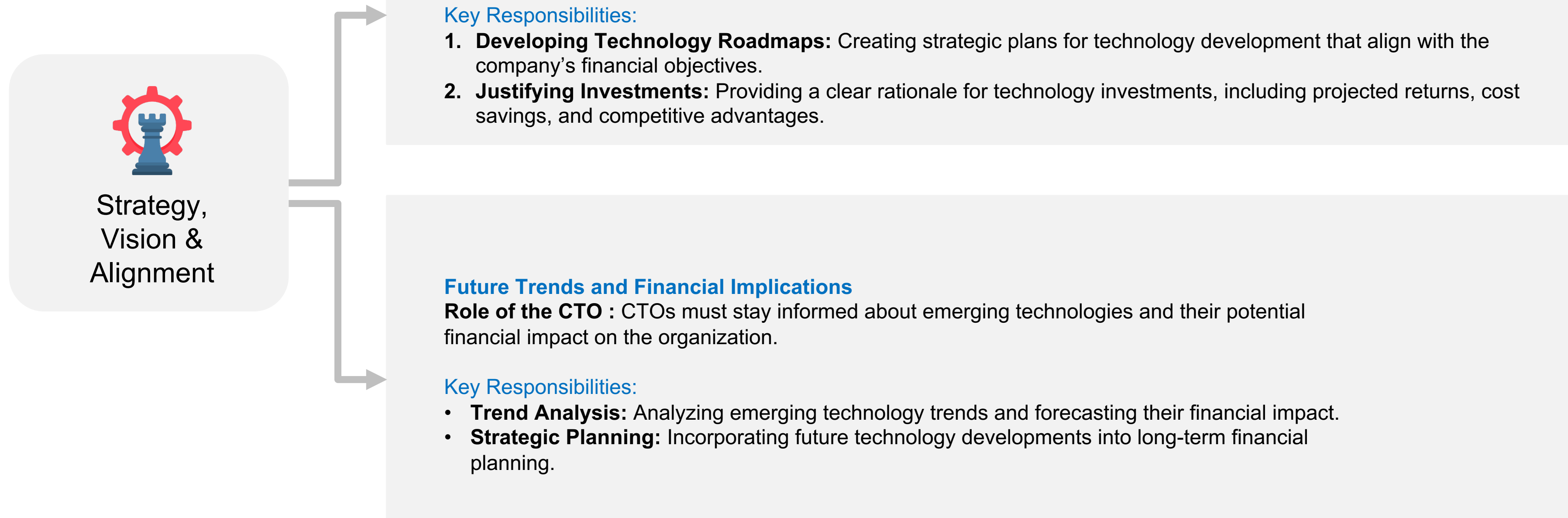
CTO's scope in financial decision-making



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CTO's scope in financial decision-making



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Evaluation /
Prioritization

Evaluating and Managing Technology Spends and Investments

Role: CTOs are responsible for assessing the financial viability of technology projects and ensuring that investments yield the expected benefits.

Key Responsibilities:

1. **ROI Analysis:** Evaluating the return on investment (ROI) for technology projects and ensuring that they contribute to financial performance.
2. **Budget Management:** Managing the technology budget, including forecasting and cost control.

Strategic Cost & RoI Management

Role of the CTO: The CTO plays a key role in managing and optimizing technology costs and maintaining high RoI to ensure efficient use of financial resources.

Key Responsibilities:

1. **Cost Control:** Implementing cost control measures and ensuring that technology expenditures are within budget.
2. **Efficiency Improvement:** Identifying opportunities to improve operational efficiency through technology.

CTO's scope in financial decision-making



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Communication
& Assignment

Communicating Financial Impact

Role: CTOs need to effectively communicate the financial impact of technology decisions to other executives and stakeholders.

Key Responsibilities:

1. **Reporting:** Providing clear and concise reports on the financial performance of technology investments.
2. **Stakeholder Engagement:** Engaging with stakeholders to explain the financial benefits and risks of technology projects.



Risk
Management

Risk Profiling and Management

Role: CTOs must identify and mitigate risks associated with technology investments, which can have significant financial implications.

Key Responsibilities:

1. **Risk Assessment:** Evaluating potential risks related to new technologies and their impact on the company's financial health.
2. **Mitigation Strategies:** Developing strategies to manage and reduce risks, such as cybersecurity threats or project failures.

CIOs about Financial Leadership



Raj Kushwaha
CTO & MD,
Warburg Pincus

The approach requires adding a rigor for thinking about technology **spend** in two distinct buckets.

The first being the **lights-on bucket**, which represents the true maintenance costs of technology and should scale with revenue, just like any other operational function. This component should be the only component part of the annual budget cycle and subject to normal cost optimization approaches.

The second bucket, hopefully the bigger bucket, should be the **technology innovation bucket**, which should be structured as a series of venture capital bets with a level of innovation funding the company is willing to make on an annual basis with an aggregate expectation of 3X quantified returns over a three-year period.

The venture funding should support a portfolio of bets (projects) that have different risk profiles and risk horizons, all structured with near-term incremental success criteria that are brutally transparent and cleanly measurable, and are governed with a venture board like governance comprised of cross-functional leaders. It should be acknowledged and celebrated that about 30 percent of the best will fail, but the balance will deliver a material and measurable ROI, which in aggregate returns 3X money for the annual innovation spend level. Key being a measurable returns requirement, just like a successful venture bet would have.

CIOs about Financial Leadership



Raj Kushwaha
CTO & MD,
Warburg Pincus

(Continued)

More progressive companies can even extend the concept by linking variable compensation of teams to venture returns with success multipliers for annual value creation. This venture approach to innovation spending not only creates a maniacal focus on technology-enabled value execution, but it also forces technologists to start thinking about financials and returns more so than they normally would. Also, it forces the discipline of six- to nine-month value creation cycles that can measurably drive variable compensation.

The venture approach to innovation spending can be transformative and represents a culture shift for organizations — from that of managing technology spending to that of technology investing — and a skill set shift for technology leaders — from being technology CIOs to venture-backed CEOs.

CIOs about Financial Leadership



Bradley Strock
CIO,
PayPal

I would say the most important lesson about IT finance is that it is business finance. What I mean by that is the **language of business is finance**, and it is fundamental to conducting business anywhere. So, while there are certainly particular nuances to IT from a financial perspective, I view all of it as business finance. Whether it is determining what projects to invest in, evaluating the efficiency of processes, or [evaluating] how technology is contributing to the growth of the company, finance is the primary tool we use in business.

What I often find with technologists is that they lack even the basics of how finance is used in a business. If the language of business is finance and you can't speak even the basics of that language, it means you are missing out on the fundamentals of the enterprise of which you are a part. Basics like the difference between operating expense and capital expense remain a mystery for many engineers.

Perhaps I'm biased because I came from a finance background very early in my career. In fact, I only half-jokingly tell people that if I hadn't become a CIO, I might have ended up as a CFO. But the truth is that having a basic grasp of core financial concepts is extremely important to being a successful technologist, especially at the leadership level.

So, take the time to invest in yourself and learn the basics. Be able to read and understand a simple Income Statement and Balance Sheet. **Know how capital spending becomes an asset and then depreciates, becoming an operating expense. Understand the difference between income and cash flow.** The aim is not to become a CPA, but to be able to be conversant in the lingua franca of business i.e. finance

CIOs about Financial Leadership

Strategic Partner:



Jamie Cutler
CIO,
Qep Resources Inc.
(US-based oil & gas major)

In my time leading technology, process and information teams in various industries, there is one lesson I have learned that continues to hold true today and will in the future. **As a leader in any organization, no matter the level or title you hold, being able to tie investment to results is the most important financial discipline you can put into place in your organization.** Corporate leaders, boards, and heads of operations discuss profit and loss, cost of goods, cost of sales, and operating expense. As a company leader, you need to speak the same language. Too often I hear IT personnel talk about IT as a “fixed” cost or as a cost of doing business. The paradigm of IT has shifted and thus as leaders we need to use the same concepts and measuring sticks as our peers across the company. The key is to hold yourself and your team accountable for results based on financial investment.

Start with tying projects to results; talk with your **executive steering committee** about the “why” of projects and the expected results from each effort, and classify projects based on the “area” of a project. Is the project for future efficiency, is it for process improvement, or is it driven by a regulatory requirement? What did the project return to the company? How do you measure return on the project? Have criteria that you have already vetted with your peers around how to measure this return (hours returned to employees, faster to market time, better consumer experience, etc.). Use both empirical, number-based data and anecdotal information from business units to measure the return; both of these criteria have a place in the discussion.

CIOs about Financial Leadership



Jamie Cutler
CIO,
Qep Resources Inc.

(Continued)

Expense investment can be more difficult to quantify on a return basis. My belief with expense is that every dollar spent in IT should show a value back to the company in some manner. Use broader more service-level conversations for expense dollar investments. What levels of service are purchased with the investment? How does this compare with “as a service” vendors and other external providers? Show how your metrics compare with those of external providers. Don’t forget security, either. Quantifying the idea of keeping your employees and customer’s information safe is an investment, and the downside, as we all know, is far greater than the investment. Keep in mind, security is not a zero-sum game; enhancing security tools and process can increase safety and increase usability. The financial industry has been successful in enhancing security while increasing usability.

The key to financial investment is to know where your IT dollars are going and then to have the dialogue across the organization about those investments in results terms, not IT terms

CIOs about Financial Leadership

Strategic Partner:



Ron Guerrier
CIO,
Farmers
Insurance

(US-based Insurance
MNC)

The most important lesson that I've learned about IT Finance comes with the [prep, polish, and presentation](#).

Most IT professionals don't think too much about the **evangelism needed to get IT initiatives funded and launched**. I've learned that taking the extra time to ensure that you've prepared all the necessary documents and all accounting treatments are true. The next learning is that the details must be polished, as the flow and logic must tell a story that resonates with those approving the request. The final step is a presentation that not only provides low-, mid-, and high-cost options with a recommendation, but also has a few mock-ups or wire frames that give a visual deliverable. On the latter, it's worthwhile to create these samples with diminishing features based on the low-, mid-, or high-option chosen. In a world of agile, proof of concepts, etc., this should be an output at inception.

Embedding a finance subject matter expert on your team will ensure the economics are sound. And if you apply the right preparation, polish, and presentation, you'll find success in getting your IT project through the finance gauntlet.

CIOs about Financial Leadership



Peter Weis
CIO,
Matson

(Global Logistics Major)

As the pace of technology change accelerates, **there's a strong tendency for CIOs to expend great energy worrying that their skill sets are constantly on the brink of obsolescence.** As I reflect on the technology waves of the previous 20 years, I think that worry is largely misplaced. Staying technologically current is in our genes. It's who we are and how we've been trained. Technology is our native language.

Conversely, the **CEO's native language isn't technology. It's finance.** A CEO's success is directly tied to the ability to improve a company's financial levers: EPS, ROIC, EBIDTA, P/E, NPV, and WACC are core elements of this language, and it's the language CIO's must master. Over time, I've learned to *largely spare them the acronyms of my native tongue. I mention SaaS, PaaS, SDN, VoIP, DLP, and IAM sparingly and only when necessary* as they relate to business issues: top line growth, margin improvements, customer experience, or risk mitigation.

Great CIOs must be viewed as "moving the needle" in the CEO's eyes. **Building fluency in the language of finance is among the fastest ways to a corner office and was also critical for me in performing a "CIO+" role.**

My financial acumen played a key role, as I was entrusted with starting up and operating a global supply-chain business unit for four years.

$$WACC = \left(\frac{E}{E + D} * Re \right) + \left(\frac{D}{E + D} * Rd * (1 - T) \right)$$

WACC is the rate that a business is expected to pay to finance its assets

CIOs about Financial Leadership

(Continued)



Peter Weis
CIO,
Matson
(Global Logistics
Major)

How does one become viewed as a needle-moving CIO? Here are five quick tips:

1. **YOU THE CTO Learn your company's financial levers and metrics** so that your **entire team understands how your company's performance is measured** by the board, investors, and outside analysts.
2. **Train your entire IT leadership (CTO – 1, 2, 3) team** so that corporate finance is baked into your team's DNA. Sell the vision and insist on commitment.
3. **Build (GET IT BUILT & USE) the best financial models in your company**. CIOs who demonstrate financial rigor when analyzing IT investments build trust and credibility. Demonstrating financial stewardship translates into greater success in competing for your company's scarce capital.
4. **Cut discretionary, soft-benefit enhancements as a way to fund real innovation**. You'll gain respect as an innovator and as a leader who is unafraid of making tough calls. **Be both a shrewd investor and a fearless cost cutter.**
5. Avoid the **"CEO squirm factor."**
 - Use **financial language wherever possible in C-level and board meetings**.
 - Avoiding needless technology jargon builds rapport and demonstrates that you too *are* the business.

While it's common for us to progress in our IT careers without mastering these skills, it's also ultimately career-limiting. IT executives who master this second language are more likely to get hired, to get promoted, and to have a seat at the executive table.

Technology skills are necessary for success, but they're not sufficient. Finance skills are seen as more-scarce and in many CEO's eyes, more valuable. I wish I'd learned this sooner.



5.3

Implementing financial strategies in IT / tech function

Balanced Score Card (BSC)



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To be covered in Session 6
(since it was not covered in the first module for Batch 5)



5.4

Building a financially savvy technology team

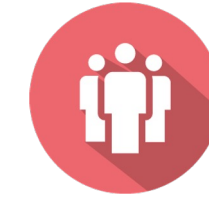
Building a financially savvy IT / technology team



Operational Scope



Org. scope



People scope

Building a financially savvy technology team is crucial for a Chief Technology Officer (CTO) who aims to align technology initiatives with business goals and optimize budget utilization.



PROCESS

TOOLS & SYSTEMS

CULTURE

LEARNING &
DEVELOPMENT

PERFORMANCE
MANAGEMENT

Building a financially savvy technology team



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Strategic Partner:

leadup
universe



Organizational
scope



Culture

Foster a Culture of Cost & RoI Awareness

Encourage a culture where team members are conscious of costs and resource utilization to maintain high RoI. Promote practices such as regular cost & RoI reviews, budgeting discipline, and efficient use of resources.

Reference: Forbes – Cost Consciousness in Engineering and Dev Teams

Encourage Cross-Functional Collaboration

Promote collaboration between technology and finance teams to ensure that technology projects are aligned with financial goals. Cross-functional teams can work together to develop budgets, forecast financial impacts, and identify cost-saving opportunities.

Promote Data-Driven Decision Making

Encourage the use of data and analytics in decision-making. Financial data should be used to assess project viability, measure performance, and make informed decisions about resource allocation.

Building a financially savvy IT / technology team



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Operational
Scope



Process

Establish Financial Goals and KPIs to every initiative

Set clear financial goals and KPIs (Key Performance Indicators) related to technology initiatives. Communicate these goals to your team and track progress regularly to ensure alignment with financial objectives.

Involve the Team in Budgeting and Financial Planning

Engage your technology team in the budgeting and financial planning processes. This involvement helps team members understand the financial implications of their projects and fosters a sense of ownership over budgetary constraints.

Tools & Systems

Implement Cost Tracking and Reporting Tools

Utilize cost tracking and reporting tools to monitor technology expenses and ensure that projects stay within budget. Regular reporting helps identify any deviations and allows for timely corrective actions.

Building a financially savvy IT / technology team



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People
scope



Learning & Development

Educate your entire team on financial basics: Ensure that your technology team understands basic financial concepts such as budgeting, cost-benefit analysis, and ROI. Financial literacy helps team members make more informed decisions about technology investments and resource allocation.

Reference Article:

Provide role-relevant advanced financial training, including of systems and tools: Invest in training programs and resources that focus on financial management and business acumen. Workshops, online courses, and mentorship from financial experts can enhance the financial skills of your technology team.



Evaluation & Recognition

Integrate Financial Metrics into Performance Evaluations Incorporate financial metrics into performance evaluations to emphasize the importance of financial considerations. Metrics might include budget adherence, cost savings from efficiency improvements, and ROI on technology projects.

Recognize and reward financial acumen: Acknowledge and reward team members who demonstrate strong financial acumen and contribute to cost-saving initiatives. Recognition can motivate others to adopt similar practices and reinforce the importance of financial responsibility.



5.4

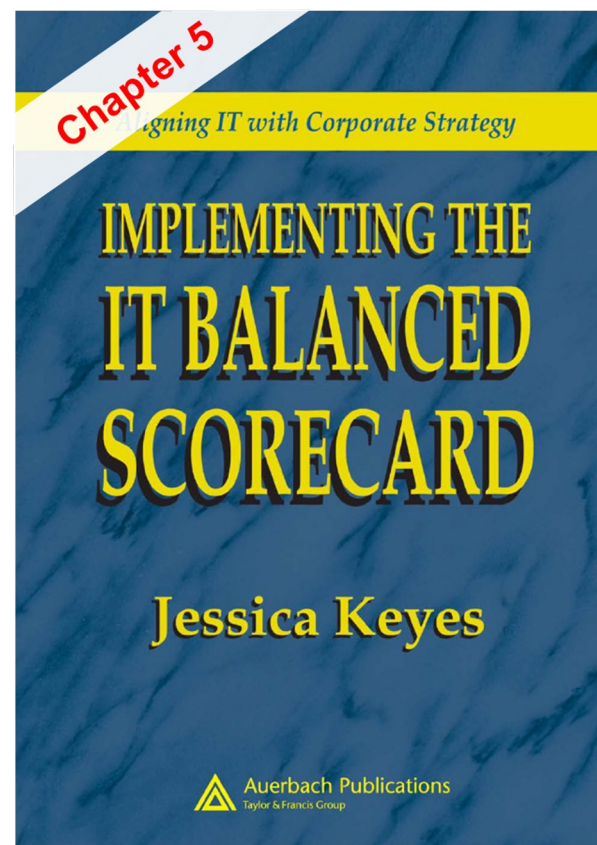
Session Reference

Session 5 reference

Strategic Partner:



SESSION'S REFERENCE (MANDATORY):

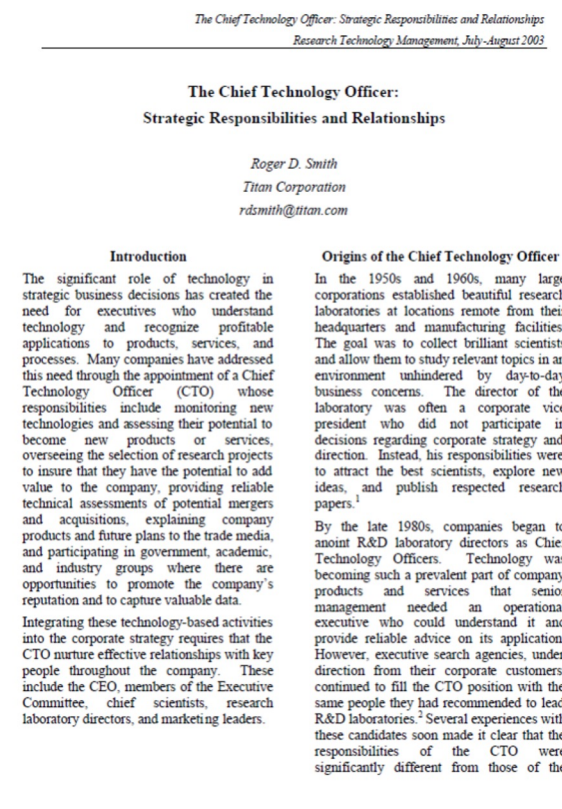


Mandatory Reading portions for our course:

Chapter 5
(the rest of the chapters of this book are optional, i.e. not required to complete assessments / assignments of this module.)

URL to view:
https://drive.google.com/file/d/1ciPvU-RzAkuy16TVQYS2VUi_sAlgn5UD/view

SESSION'S REFERENCE (OPTIONAL):



URL to view:
<https://rb.gy/ggoqrs>

Thank You